
IEEE Access - Decision on Manuscript ID Access-2025-52745

1 개의 메일

IEEE Access <onbehalf@manuscriptcentral.com>

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14-Dec-2025

Dear Mr. KANG:

I am writing to you regarding manuscript # Access-2025-52745 entitled "Topology-Constrained NeuroB-Rep: Contact-Aware CAD Reconstruction from 3D Point Clouds" which you submitted to IEEE Access.

Your article was peer reviewed with interest but has not been recommended for publication in its current form. We strongly encourage you to address the reviewers' concerns, which can be found at the bottom of this letter, and resubmit your article to IEEE Access once you have updated it accordingly.

Please note that IEEE Access has a binary peer review process. Therefore, to uphold quality to IEEE standards, an article is rejected even if it requires minor edits.

When updating your manuscript, you should elaborate on your points and clarify with references, examples, data, etc. If you disagree with any technical points the reviewers have made, please include your counterarguments in your response to the reviewers (more information detailed below) and work this into the updated manuscript.

Also, note that if a reviewer suggested references, you should only add those that are relevant to your work if you feel they strengthen your article. Recommending references to specific publications is not appropriate for reviewers and you should report excessive cases to ieeeaccessEIC@ieee.org. Authors are not obligated to cite articles that are recommended by the reviewers, and the final decision on the article will not be influenced by whether or not authors cite these suggested references.

IEEE Access allows one opportunity to resubmit. If the updated manuscript is determined not to have addressed all of the previous reviewers' concerns, or if the Associate Editor still has substantial technical concerns, the article will be rejected and no further resubmissions will be allowed.

When you are ready to resubmit your updated article, you can do so in the [IEEE Author Portal](#). When you log into the IEEE Author Portal you will see the title of the rejected article and the option to "Start Resubmission".

Upon resubmission you will be asked to upload the following 3 files:

1) A document containing your response to reviewers from the previous peer review. The "response to reviewers" document (template attached) should have the following regarding each comment: a) Reviewer's concern, b) your response to the concern, c) your action to remedy the concern. **The document should be uploaded with your manuscript files under "Author's Response Files."**

2) Your updated manuscript with all your individual changes highlighted, including grammatical changes (e.g. preferably with the yellow highlight tool within the pdf file). This file should be uploaded with your manuscript files as "*Highlighted PDF*."

3) A clean copy of the final manuscript (without highlighted changes) submitted as a Word or LaTeX file, and as a PDF, both submitted as the "*Main Manuscript*."

****IMPORTANT:** Please see the attached Resubmission Checklist that details all the items listed above. Please utilize this checklist to ensure you have made the necessary edits to your manuscript, and to ensure you have all the necessary files prepared prior to resubmission.

***** AUTHOR LIST CHANGES:** If your revised manuscript has an updated author list, you will need to submit a formal request to the Editor by completing the attachment labelled 'Request for Byline Change,' and uploading it as '*Request for byline change form*.' This should include a DETAILED justification explaining each author's contribution(s) to the work. You will also need to provide the justification for the author change during the submission process. Change in

the author list is considered rare and exceptional, and the decision to allow such changes rests with the Editor. Once the list and order of authors has been established, the list and order of authors should not be altered without permission of all living authors of that article.

We sincerely hope you will update your manuscript and resubmit soon. Please contact me if you have any questions.

Thank you for your interest in IEEE Access.

Sincerely,

Dr. Jiju Poovvancheri
Associate Editor, IEEE Access
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Reviewers' Comments to Author:

Reviewer: 1

Comments:

Major Concerns

1. Clarity of Contribution vs. Point2CAD

- The paper positions itself as an extension of Point2CAD [2], but the description of what exactly is reused (e.g., segmentation, hybrid representation) vs. what is novel (contact graph, gated optimization) could be clearer in the Introduction and Method sections.
- Suggest: Add a concise table or flowchart comparing stages of Point2CAD and NeuroB-Rep to highlight the added components.

2. Evaluation Protocol Consistency

- While the hybrid evaluation (mm for geometry, normalized scale for contacts) is justified, it may confuse readers. A clear statement should be added in Section IV-B that this is intentional to decouple geometric accuracy from topological closure.
- Suggest: Include a short subsection "IV-B3 Rationale for Hybrid Evaluation" explaining the design choice.

3. Missing Baseline Comparison on Key Metrics

- Table 3 reports results only for the proposed method. A direct comparison with Point2CAD on the same mm-geometry metrics (Chamfer, Hausdorff, etc.) is missing.
- Suggest: Add a column in Table 3 or a new table comparing NeuroB-Rep vs. Point2CAD on all geometric accuracy metrics under the same ICP-aligned protocol.

4. Limited Discussion on Failure Cases

- Figures 5 and 6 show layer misassignment, but the text does not analyze why this occurs or how the method could be improved to handle such cases.
- Suggest: Expand Section VII to include a paragraph on failure modes and potential mitigations (e.g., better boundary cues, adaptive layer thresholds).

Minor Issues

1. Abstract Wording

- The abstract claims "geometric errors on the order of 0.01 mm," but Table 3 shows some errors (e.g., Height error for impeller) up to 0.023 mm. The wording should be adjusted to reflect the range of errors.
- Suggest: Change to "geometric errors within 0.01–0.03 mm" or similar.

2. Formula Notation Consistency

- In Eq. (2), d_{ij}, k_{dij}, k is used, but later s_{ij}, k_{sij}, k appears in Eq. (3) without a clear link. Ensure all distance symbols are defined consistently.
- Suggest: Add a sentence after Eq. (2) linking d_{ij}, k_{dij}, k and s_{ij}, k_{sij}, k .

3. Figure References in Text

- Some figures are referenced out of order (e.g., Fig. 5 appears before Fig. 4 in the narrative). Ensure figures are cited in numerical order.
- Suggest: Reorder citations or adjust figure numbering.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

1) Does the paper contribute to the body of knowledge?: yes.

2) Is the paper technically sound?: yes.

3) Is the subject matter presented in a comprehensive manner?: yes.

4) Are the references provided applicable and sufficient?: yes.

5) Are there references that are not appropriate for the topic being discussed?: No

5a) If yes, then please indicate which references should be removed.:

Reviewer: 2

Comments:

The authors propose a Topology-Constrained Neural Boundary Representations (NeuroB-reps) solver that couples a hybrid surface model with a topology-aware objective and a stable contact gating schedule. The surface model combines analytic primitives for simple patches and neural implicit signed distance functions for freeform regions. The objective penalizes gaps between surfaces, G1/G2 discontinuities, and self-intersections while regularizing the coaxiality and angular consistency of primitives. This novel approach is tested on three industrial models (shaft, impeller, and casing), demonstrating sub-millimeter level accuracy.

Despite the valuable results, numerous issues remain regarding paper structure, reproducibility, level of detail, and positioning with respect to the literature.

Comment 1:

The authors present and compare their work only with Point2CAD. The authors are advised to include all of the following papers, comment on them, and, if code is available, compare their results with those produced by the following papers, thereby extending the related work section. Additionally, it is expected that Table 3 be extended to include accuracy outcomes from all the methods below:

Li, L., Sung, M., Dubrovina, A., Yi, L., Guibas, L.J., 2019. Supervised Fitting of Geometric Primitives to 3D Point Clouds. *Comput. Vis. Pattern Recognit.* <https://doi.org/10.1109/cvpr.2019.00276>

Guo, H., Liu, S., Pan, H., Liu, Y., Tong, X., Guo, B., 2022. ComplexGen. *ACM Trans. Graph.* <https://doi.org/10.1145/3528223.3530078>

Huang, J., Zhang, Y., Sun, M., 2021. PrimitiveNet: Primitive Instance Segmentation with Local Primitive Embedding under Adversarial Metric. *IEEE Int. Conf. Comput. Vis.* <https://doi.org/10.1109/iccv48922.2021.01506>

Liu, J., 2020. An adaptive process of reverse engineering from point clouds to CAD models. *Int. J. Comput. Integr. Manuf.* <https://doi.org/10.1080/0951192x.2020.1803501>

Liu, Y., Obukhov, A., Wegner, J.D., Schindler, K., 2023. Point2CAD: Reverse Engineering CAD Models from 3D Point Clouds. *Comput. Vis. Pattern Recognit.* <https://doi.org/10.1109/cvpr52733.2024.00361>

Maurell, I.P., Corçaque, P.L., Froes, C.L., Lemos, J.F.S.S., Oliveira, F.G., Drews, P., 2024. Geometric Deep Learning in Industrial Scenes: A Large-Scale 3D Synthetic Dataset. *Lect. notes Comput. Sci.* https://doi.org/10.1007/978-3-031-78495-8_1

Comment 2:

The introduction should include an outline of the paper structure. This is currently missing.

Comment 3:

The following paragraph from the introduction is out of place and should be moved to the earlier parts of the introduction relevant to motivation and background:

"In sectors such as the petrochemical and power generation industries, timely maintenance and replacement of critical mechanical components are of paramount importance [16], [25]. However, key components such as industrial compressors are often sourced from a limited number of overseas manufacturers (e.g., Japan, Germany, United States). This reliance introduces high costs, extended lead times, and vulnerability to global supply-chain disruptions, weakening domestic technological competitiveness over time [16]. In our industrial study, the resulting Cloud2CAD pipeline automatically converts 3D point-cloud data into high-precision CAD models, achieving average geometric errors on the order of 10^{-2} mm relative to the scan on the reported case and indicating potential reductions in design turnaround and cost in our internal evaluation."

Comment 4:

The related work section is insufficient. It should be augmented by presenting the downsides of existing methods and positioning the proposed approach with respect to the literature.

Comment 5:

Section II contains only one subsection. The text of this subsection should be integrated into the main body of Section II.

Comment 6:

Please explain: "Patches form a B-rep candidate graph with face–edge–vertex incidence; edges at patch interfaces are potential contacts."

The input consists only of points. How are faces, vertices, and edges recovered?

Comment 7:

There is only one subsection titled "1) Hybrid Fitting with INRs and Primitives." Please avoid having only one subsection.

Comment 8:

In Section III-B, it is mentioned "with instant mesh repair principles [12]."

This requires further explanation. Even though a reference is provided, this is a crucial part of the process. Create a new section named "Preliminaries" to describe existing methods.

Comment 9:

In Section III-B, the following passage lacks clear sentence structure:

"Let $\Pi_j(\cdot)$ be the projection operator onto patch j (analytic projection for primitives; gradient-based projection for the INR SDF), and define symmetric distances $d_{ij} = \|p - \Pi_j(p)\|$ and $d_{ji} = \|q - \Pi_i(q)\|$ for paired samples (p, q) . With the hinge $h_e(d) = \max(0, d - \epsilon)^2$, our gap term is"

The passage contains sentence fragments and poor expression. Please revise for clarity.

Comment 10:

"Then the self-intersection suppressor is a soft barrier" → What is a soft barrier? Please define this term.

Comment 11:

Section III-C uses subsection labels starting with "a:" (e.g., "a: Gate sets and schedules"). The paper structure should be improved to use consistent numbering.

Comment 12:

Figure 3 references panels a, b, c, and d, but only three unnumbered panels are shown.

Comment 13:

"We normalize scans to a canonical pose."

What is the purpose of this alignment? What are the effects of non-alignment? Please present evidence.

Comment 14:

"We establish a world frame by the GT-CAD datum scheme (ISO 5459)."

The authors should provide further explanation of this approach.

Comment 15:

Section IV-C states: "learning rate of 10^{-2} on a GPU device."

Since the authors are minimizing Equation (1), how was this minimization implemented on GPUs? Please clarify the computational approach.

Comment 16:

Section VI: Can the authors provide an artifact URL and confirm public availability?

Comment 17:

Table 3 should contain actual measurements and comparisons with Point2CAD and other algorithms.

Comment 18:

Peculiar terminology: "Reproducibility hygiene. For reproducibility, we advise pinning the environment (hash seeds, dependency lockfile)."

Please correct and align terminology with standard conventions in the computer vision and CAD literature.

Additional Questions:

Please confirm that you have reviewed all relevant files, including supplementary files and any author response files, which can be found in the "View Author's Response" link above (author responses will only appear for resubmissions):
Yes, all files have been reviewed

1) Does the paper contribute to the body of knowledge?: Marginally. Contribution is not clear with respect to State of the Art

2) Is the paper technically sound?: Yes

3) Is the subject matter presented in a comprehensive manner?: Not at all. Details are provided in the comments.

4) Are the references provided applicable and sufficient?: Yes.

5) Are there references that are not appropriate for the topic being discussed?: No

5a) If yes, then please indicate which references should be removed.:

If you have any questions, please contact article administrator: Dr. Ratnesh Bajpai r.k.bajpai@ieee.org



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